
Unit – 1 : Session – 8 : SLO – 1 | Kernel Data Structures — Symbol Code Puzzle

I. Secret Message

1. Decoding the symbols with the key at the bottom of the page reveals:
2. "KERNEL DATA STRUCTURES KEEP TRACK OF PROCESSES, MEMORY, FILES AND DEVICES."

II. What Are Kernel Data Structures?

1. PROCESS LIST / PCB TABLE — one Process Control Block per process (state, registers, PC).
2. READY QUEUE and WAIT (device) QUEUES — linked lists of process descriptors used by schedulers.
3. MEMORY MAPS / PAGE TABLES — track free and allocated frames and address translation.
4. OPEN FILE TABLE and INODE structures — track files opened by processes.
5. DEVICE TABLE and I/O request queues — track devices and pending operations.